Vortex Tech Internship – Week 2

Hands-On with Basic Security Tools

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Track: Cyber Security

Introduction

This week's task focused on moving from cybersecurity theory into practical work by developing a basic password strength evaluator in Python and performing a local port scan using Nmap. Both activities were conducted safely on my own machine.

Task 1 — Password Strength Evaluator

Objective:

The objective of this task was to create a Python script that evaluates password strength based on length, character variety, and common-password checks.

Tools Used:

Python and Visual Studio Code

Code:

password_checker.py

C: > Users > hp > Desktop > password_checker.py

```python
COMMON_PASSWORDS = [
    "123456",
    "password",
    "qwerty",
    "12345678",
    "admin",
    "letmein"
]


def check_password_strength(password):
    feedback = []

    if password.lower() in COMMON_PASSWORDS:
        return "Very Weak", ["This is a very common password. Choose a unique password."]

    length_ok = len(password) >= 8
    uppercase_ok = any(c.isupper() for c in password)
    lowercase_ok = any(c.islower() for c in password)
    number_ok = any(c.isdigit() for c in password)
    special_ok = any(not c.isalnum() for c in password)

    checks_passed = sum([
        length_ok,
        uppercase_ok,
        lowercase_ok,
        number_ok,
        special_ok
    ])

    if not length_ok:
        feedback.append("Use at least 8 characters.")

    if not uppercase_ok:
        feedback.append("Add at least one uppercase letter.")

    if not lowercase_ok:
        feedback.append("Add at least one lowercase letter.")

    if not number_ok:
        feedback.append("Add at least one number.")
```

```python
    if not special_ok:
        feedback.append("Add at least one special character.")

    if checks_passed <= 2:
        strength = "Weak"
    elif checks_passed <= 4:
        strength = "Medium"
    else:
        strength = "Strong"

    if not feedback:
        feedback.append("Good password structure.")

    return strength, feedback


def main():
    password = input("Enter a password to evaluate: ")

    strength, feedback = check_password_strength(password)

    print("\nPassword Strength:", strength)
    print("Feedback:")

    for message in feedback:
        print("-", message)


if __name__ == "__main__":
    main()
```

Output:

Weak Password Test

```
Problems   Output   Debug Console   Terminal   Ports

PS D:\Microsoft VS Code> & "C:\Program Files\Python313\python.exe" c:/Users/hp/Desktop/password_checker.py
Enter a password to evaluate: cyber

Password Strength: Weak
Feedback:
- Use at least 8 characters.
- Add at least one uppercase letter.
- Add at least one number.
- Add at least one special character.
```

Medium Password Test

```
Enter a password to evaluate: pak12345

Password Strength: Medium
Feedback:
- Add at least one uppercase letter.
- Add at least one special character.
```

Strong Password Test

```
Enter a password to evaluate: Vortex@4312

Password Strength: Strong
Feedback:
- Good password structure.
```

Task 2 — Nmap Local Port Scan

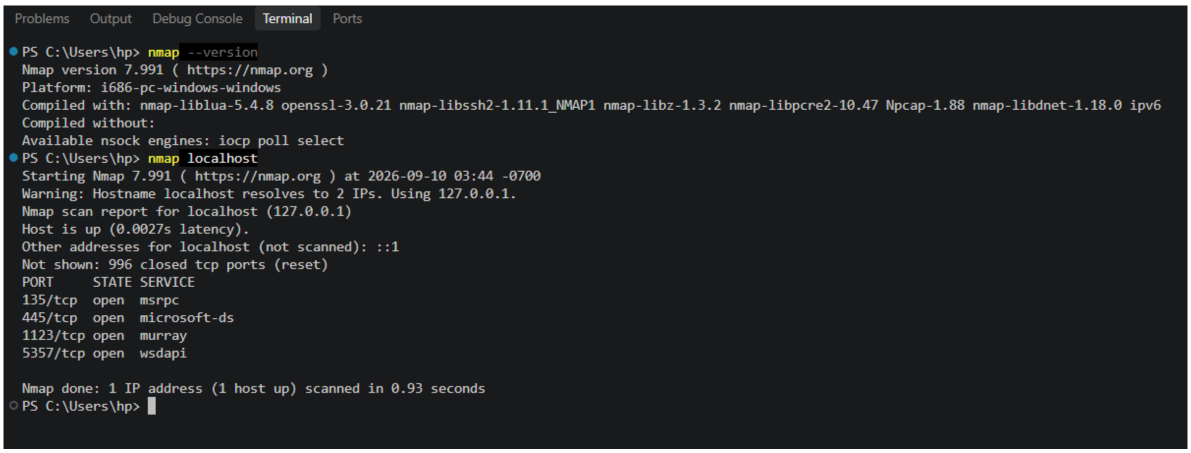
Objective:

The objective of this task was to use Nmap to scan my own computer and identify open TCP ports and the services associated with them.

Command Used:

nmap localhost

Nmap Scan Output:



Port Scan Results:

Port	State	Service Reported by Nmap
135/tcp	Open	msrpc
445/tcp	Open	microsoft-ds
1123/tcp	Open	murray
5357/tcp	Open	wsdapi

Reflection

This task made me realize that even simple things like a weak password or an open port can cause security problems. A common or easy-to-guess password can make it easier for someone to get into an account. The password checker showed me that using a longer password with a mix of uppercase and lowercase letters, numbers, and special characters can make it harder to guess.

The Nmap scan also helped me understand that open ports mean that certain services are running and accessible on a system. If a service is not needed or is not properly secured, it can give an attacker another possible way to target the system.

Overall, this task gave me a better understanding of how small security weaknesses can increase the risk to a system and why basic security checks are important.